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System and method for charging vehicle's higher-voltage power supply using lower-voltage charging station

Abstract

An apparatus includes an AC input configured to receive an AC electrical signal and a DC input configured to receive a first DC electrical signal having a lower voltage and a second DC electrical signal having a higher voltage. The apparatus also includes an on-board charger configured to charge a power supply using the AC electrical signal. The on-board charger includes an AC/DC PFC converter and a DC/DC converter. The apparatus further includes switches configured to form and break electrical connections between the AC input, DC input, AC/DC PFC converter, DC/DC converter, and power supply. In addition, the apparatus includes a controller configured to adjust the electrical connections formed by the switches. The controller is configured, when the DC input receives the first DC electrical signal, to place the switches in a configuration to couple the DC input to the PFC converter and couple the PFC converter to the power supply.

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Background/Summary

TECHNICAL FIELD

(1) This disclosure relates generally to electric vehicles. More specifically, this disclosure relates to a system and method for charging a vehicle's higher-voltage power supply using a lower-voltage

charging station.

BACKGROUND

(2) An electric vehicle includes a power supply (such as one or more battery modules) that provides electrical power to one or more electric motors and other components of the electric vehicle. The power supplies being used in electric vehicles have been generally increasing in voltage level over time. For example, earlier electric vehicles often used power supplies that provided around 400 volts, and the current trend is to use power supplies that provide around 800 volts. It is expected that this trend will continue and that even higher-voltage power supplies will be developed and used in electric vehicles.

SUMMARY

(3) This disclosure relates to a system and method for charging a vehicle's higher-voltage power supply using a lower-voltage charging station.

(4) In a first embodiment, an apparatus includes an alternating current (AC) input configured to receive an AC electrical signal and a direct current (DC) input configured to receive a first DC electrical signal having a lower voltage and a second DC electrical signal having a higher voltage. The apparatus also includes an on-board charger configured to charge a power supply using the AC electrical signal, where the on-board charger includes an AC/DC power factor correction (PFC) converter and a DC/DC converter. The apparatus further includes switches configured to selectively form and break electrical connections between the AC input, the DC input, the AC/DC PFC converter, the DC/DC converter, and the power supply. In addition, the apparatus includes a controller configured to adjust the electrical connections formed by the switches. The controller is configured, when the DC input receives the first DC electrical signal having the lower voltage, to place the switches in a first configuration to couple the DC input to the AC/DC PFC converter and couple the AC/DC PFC converter to the power supply.

(5) In a second embodiment, an electric vehicle includes at least one electric motor and a power supply configured to provide electrical energy to the at least one electric motor. The electric vehicle also includes an AC input configured to receive an AC electrical signal and a DC input configured to receive a first DC electrical signal having a lower voltage and a second DC electrical signal having a higher voltage. The electric vehicle further includes an on-board charger configured to charge the power supply using the AC electrical signal, where the on-board charger includes an AC/DC PFC converter and a DC/DC converter. The electric vehicle also includes switches configured to selectively form and break electrical connections between the AC input, the DC input, the AC/DC PFC converter, the DC/DC converter, and the power supply. In addition, the electric vehicle includes a controller configured to adjust the electrical connections formed by the switches. The controller is configured, when the DC input receives the first DC electrical signal having the lower voltage, to place the switches in a first configuration to couple the DC input to the AC/DC PFC converter and couple the AC/DC PFC converter to the power supply.

(6) In a third embodiment, a method includes determining whether an AC input is receiving an AC electrical signal, a DC input is receiving a first DC electrical signal having a lower voltage, or the DC input is receiving a second DC electrical signal having a higher voltage. The method also includes, based on the determination, controlling switches to selectively form and break electrical connections between the AC input, the DC input, an AC/DC PFC converter of an on-board charger, a DC/DC converter of the on-board charger, and a power supply. Controlling the switches includes, when the DC input receives the first DC electrical signal having the lower voltage, placing the switches in a first configuration to couple the DC input to the AC/DC PFC converter and couple the AC/DC PFC converter to the power supply.

(7) Other technical features may be readily apparent to one skilled in the art from the following figures, descriptions, and claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) For a more complete understanding of this disclosure and its advantages, reference is now made to the following description taken in conjunction with the accompanying drawings, in which like reference numerals represent like parts:
- (2) FIG. 1 illustrates an example vehicle supporting charging of the vehicle's higher-voltage power supply using a lower-voltage charging station according to this disclosure;
- (3) FIGS. 2 through 4 illustrate example configurations of a vehicle during charging of the vehicle's power supply using different sources of electrical power according to this disclosure;
- (4) FIG. 5 illustrates a specific example implementation of a vehicle supporting charging of the vehicle's higher-voltage power supply using a lower-voltage charging station according to this disclosure;
- (5) FIG. 6 illustrates an example control system for controlling charging of a vehicle's power supply according to this disclosure;
- (6) FIG. 7 illustrates another specific example implementation of a vehicle supporting charging of the vehicle's higher-voltage power supply using a lower-voltage charging station according to this disclosure; and
- (7) FIG. 8 illustrates an example method for charging a vehicle's power supply according to this disclosure.

DETAILED DESCRIPTION

- (8) FIGS. 1 through 8, described below, and the various embodiments used to describe the principles of this disclosure are by way of illustration only and should not be construed in any way to limit the scope of this disclosure. Those skilled in the art will understand that the principles of this disclosure may be implemented in any type of suitably arranged device or system.
- (9) As noted above, an electric vehicle includes a power supply (such as one or more battery modules) that provides electrical power to one or more electric motors and other components of the electric vehicle. The power supplies being used in electric vehicles have been generally increasing in voltage level over time. For example, earlier electric vehicles often used power supplies that provided around 400 volts, and the current trend is to use power supplies that provide around 800 volts. It is expected that this trend will continue and that even higher-voltage power supplies will be developed and used in electric vehicles.
- (10) Increasing the voltage level of a power supply typically decreases the amount of electrical current that is needed to provide power to the components of an electric vehicle. Lowering the amount of electrical current used in an electric vehicle can provide various advantages. For instance, lower electrical currents typically suffer from smaller losses when being transported over copper wiring or other wiring, and smaller electrical losses can result in improved powertrain system efficiency and increase the driving range of an electric vehicle. Lower electrical currents can also reduce the amount of copper or other materials used in an electric vehicle's wiring, which can reduce the weight and cost of the electric vehicle.
- (11) Despite the desire to use higher-voltage power supplies in electric vehicles, a major problem exists with direct current (DC) charging stations that are used around the world to charge electric vehicles' power supplies. More specifically, DC fast charging (DCFC) stations are used at numerous locations to allow users to rapidly charge their electric vehicles' power supplies. Most DCFC stations installed across the globe are currently designed to charge electric vehicle power supplies to around 400 volts. A 400-volt charging station cannot directly charge a higher-voltage power supply, such as a power supply designed to operate at 800 volts. This is because the higher-voltage power supply would have a higher voltage than the charging station, which would prevent an electrical current from flowing from the charging station to the electric vehicle's power supply.

(12) Various approaches have been proposed to overcome these problems, but these approaches can suffer from various shortcomings. For example, one approach divides a higher-voltage power supply into multiple lower-voltage power supplies coupled in series. When an electric vehicle is coupled to a lower-voltage charging station, the lower-voltage power supplies are electrically separated from each other by breaking their series connection and are electrically coupled in parallel to the lower-voltage charging station. This approach allows the lower-voltage charging station to charge the lower-voltage power supplies since each lower-voltage power supply is independently connected to the charging station. However, this approach requires the use of separate voltage bus bars and additional junction boxes to house various components, which can increase the size, weight, and cost of the electric vehicle. Moreover, an electrical switch is used to selectively break the series connection between the lower-voltage power supplies during charging by a lower-voltage charging station. If that electrical switch fails and remains closed when it should be opened, this can create a catastrophic fault that may lead to severe damage to the electric vehicle or even injury or death to any occupants. In addition, components of the electric vehicle that are designed to use a higher voltage during operation (such as heating, air conditioning, and ventilation or “HVAC” compressors or heaters) may operate poorly or not at all when the electric vehicle's power supply is being charged using a lower-voltage charging station. This may, for instance, prevent usage of the electric vehicle's air conditioning during summer or the electric vehicle's heater during winter. While it is possible to provide a boost converter that increases the voltage provided to these components when the electric vehicle's power supply is being charged using a lower-voltage charging station, this again increases the size, weight, and cost of the electric vehicle.

(13) Another approach involves the use of an extra boost converter that can be selectively switched into and out of the path of an electrical current being provided to an electric vehicle's power supply during charging. The boost converter can be switched out of the path when the electric vehicle's power supply is being charged by a higher-voltage charging station, and the boost converter can be switched into the path when the electric vehicle's power supply is being charged by a lower-voltage charging station. However, the extra boost converter can be relatively large and heavy, which increases the size, weight, and cost of the electric vehicle. Also, if the electric vehicle is used in an area where there are few or no lower-voltage charging stations, the electric vehicle is forced to continue carrying the extra boost converter even though the extra boost converter is rarely or never used.

(14) Still another approach involves using an inverter and a motor winding of an electric motor in an electric vehicle to provide boost functionality when the electric vehicle is coupled to a lower-voltage charging station. While this approach can reduce or eliminate the need for an extra stand-alone boost converter, this approach may require major changes to existing electric motors used in electric vehicles. For example, this approach generally requires that the electric motor used in an electric vehicle make its neutral electrical lead available for use outside of the electric motor. This is typically not a standard design feature of electric motors used in electric vehicles and can lead to manufacturing difficulties. Moreover, the electric motors used in electric vehicles are not designed primarily to perform a voltage boost function, so the electric motors are not optimized to perform voltage boosting. Because of this, electric motors generally do not perform voltage boosting with high efficiencies. For instance, DC flux in a motor core can introduce challenges, such as core saturation, which increase system losses and reduce system performance. In addition, this approach may require the use of an additional junction box, which can increase the size, weight, and cost of the electric vehicle.

(15) This disclosure provides a system and method for charging a vehicle's higher-voltage power supply using a lower-voltage charging station. As described in more detail below, the approaches described in this patent disclosure use one or more components of a vehicle's hardware designed for alternating current (AC) charging in order to charge the vehicle's higher-voltage power supply if and when the vehicle is coupled to a lower-voltage DC charging station. For example, the vehicle's

hardware that is designed for AC charging is often referred to as an on-board charger (OBC), and the on-board charger often includes an AC/DC power factor correction (PFC) converter and a DC/DC converter that is used for changing the voltage level of a DC voltage produced from an AC input. The described approaches allow the PFC converter of the on-board charger to be used when the vehicle is coupled to a lower-voltage charging station, and the DC/DC converter of the on-board charger can be bypassed when the PFC converter is being used in this manner.

(16) The approaches described in this patent disclosure support the use of electrical switches to controllably reconfigure the electrical pathways in the vehicle. The vehicle's electrical pathways can therefore be configured in different ways depending on how the vehicle's power supply is being charged. For example, when the vehicle's power supply is being charged using an AC power source, the vehicle's electrical pathways can be configured to allow electrical energy from an AC input to flow through the vehicle's on-board charger (including both the AC/DC PFC converter and the DC/DC converter) in order to charge the vehicle's power supply. In this mode, the AC/DC PFC converter can be used to provide power factor correction. When the vehicle's power supply is being charged using a higher-voltage DC charging station, the vehicle's electrical pathways can be configured to allow electrical energy from a DC input to flow into the vehicle's power supply (without flowing through the on-board charger) in order to charge the vehicle's power supply. When the vehicle's power supply is being charged using a lower-voltage DC charging station, the vehicle's electrical pathways can be configured to allow electrical energy from the DC input to flow through the AC/DC PFC converter of the vehicle's on-board charger (but not through the DC/DC converter of the vehicle's on-board charger) in order to charge the vehicle's power supply. In this mode, the AC/DC PFC converter can be used to provide voltage boosting.

(17) In this way, these approaches allow for charging of a vehicle's higher-voltage power supply under various scenarios, including when the vehicle's power supply is being charged using a lower-voltage charging station. This can be accomplished using the boost functionality provided by the vehicle's on-board charger, rather than using a stand-alone boost converter or an inverter and motor winding, when the vehicle's power supply is being charged using a lower-voltage charging station. These approaches can therefore achieve reduced size, weight, and cost compared to various prior approaches. Also, the PFC converter in the vehicle's on-board charger is typically designed to achieve high efficiencies, so the PFC converter in the vehicle's on-board charger is generally much more efficient in providing boost functionality compared to using an inverter and a motor winding of an electric motor. Further, these approaches can avoid the need to have access to an electric motor's neutral lead, which can avoid associated manufacturing difficulties. In addition, the PFC converter in the vehicle's on-board charger can be used for both AC charging and lower-voltage DC charging of the vehicle's power supply, which can increase the utilization rate of the PFC converter in the vehicle's on-board charger.

(18) FIG. 1 illustrates an example vehicle **100** supporting charging of the vehicle's higher-voltage power supply using a lower-voltage charging station according to this disclosure. In this particular example, the vehicle **100** takes the form of an automotive vehicle, such as an electric or other passenger vehicle. However, any other suitable electric vehicle may be used here. Also, in this particular example, the vehicle **100** has the form of a passenger van, but other forms may be used by the vehicle **100**, such as a sedan, pickup truck, sport utility vehicle, box van, or semi tractor form.

(19) As shown in FIG. 1, the vehicle **100** includes at least one high-voltage power supply **102**. The power supply **102** is generally used to provide DC electrical power to other components of the vehicle **100**. For example, the power supply **102** may be used to provide electrical power to one or more electric motors that can be used to move the vehicle **100** forward or backward, one or more HVAC system components (such as a compressor or heater and blower), and various electrical accessories in the vehicle **100**. The power supply **102** includes any suitable source of electrical power that can provide electrical energy at a relatively high voltage, such as an 800-volt battery

module or other 800-volt nominal power supply.

(20) In this example, the power supply **102** can be charged in various ways. For example, the power supply **102** may be charged using an AC electrical signal received via an AC input **104**. The AC input **104** generally represents a connector that can be coupled to an AC power source **128**. For example, the AC input **104** may represent a connector that can be coupled via an electrical cable to receive power from an AC power grid. In some cases, the AC input **104** is provided in the vehicle **100** in order to allow the vehicle **100** to be coupled to an AC outlet in a user's home or other location for recharging of the power supply **102**. The power supply **102** may also be charged using a DC electrical signal received via a DCFC input **106**. The DCFC input **106** generally represents a connector that can be coupled to a DC power source. For instance, the DCFC input **106** may represent a connector that can be coupled via an electrical cable to receive power from a charging station **130**. In some cases, the DCFC input **106** is provided in the vehicle **100** in order to allow the vehicle **100** to be coupled to a DC charging station (either in the user's home or other location) for rapid recharging of the power supply **102**. Each input **104**, **106** includes any suitable structure configured to be coupled via an electrical cable or other mechanism to a suitable power source.

(21) The vehicle **100** also includes an on-board charger **108**, which is normally used for converting the AC electrical signal received via the AC input **104** into a suitable DC voltage (which is used as described below to support charging of the power supply **102** by a lower-voltage charging station **130** in accordance with this disclosure). The on-board charger **108** here includes a cascaded pair of converters, namely an AC/DC PFC converter **110** and a DC/DC converter **112**. When the power supply **102** is being recharged using an AC power source **128**, the AC/DC PFC converter **110** generally operates to convert an AC electrical signal into a DC electrical signal. The AC/DC PFC converter **110** performs this function while increasing the voltage level of the DC electrical signal relative to the voltage level of the AC electrical signal (thereby providing a voltage “boost”). In addition, the AC/DC PFC converter **110** performs this function while providing power factor correction. Power factor correction refers to operation that increases the power factor of a load (the vehicle **100**), which helps to improve the efficiency of the AC distribution system to which the vehicle **100** is attached. Since the PFC converter **110** can perform power factor correction while boosting the voltage, the PFC converter **110** is often referred to as a boost PFC converter. The AC/DC PFC converter **110** includes any suitable structure configured to convert AC electrical energy into DC electrical energy while increasing the voltage of the DC electrical energy and providing power factor correction.

(22) The DC/DC converter **112** generally operates to convert a DC output voltage produced by the AC/DC PFC converter **110** into a suitable DC voltage for recharging the power supply **102**. As a result, the DC/DC converter **112** regulates the DC output voltage from the AC/DC PFC converter **110** so that the regulated DC voltage is suitable for use in recharging the power supply **102**. The DC/DC converter **112** can also operate to electrically isolate the power supply **102** from the AC/DC PFC converter **110**. In some embodiments, the DC/DC converter **112** may provide a DC voltage that is around 800 volts to the power supply **102**. The DC/DC converter **112** includes any suitable structure configured to convert one DC voltage into another DC voltage.

(23) The vehicle **100** further includes various switches **114-124**, which are used to reconfigure the electrical pathways in the vehicle **100** to support various charging modes for the power supply **102**. More specifically, the switches **114-124** here can be used to permit the power supply **102** to be recharged using an AC electrical signal received via the AC input **104**, a higher-voltage DC electrical signal received via the DCFC input **106**, or a lower-voltage DC electrical signal received via the DCFC input **106**. Note that the phrase “lower-voltage” is used in this patent disclosure to refer to a DC voltage that is lower than the nominal DC voltage provided by the power supply **102**, and the phrase “higher-voltage” is used in this patent disclosure to refer to a DC voltage that is higher than the voltage of the power supply **102** (at least prior to charging of the power supply **102** using the DC electrical signal received via the DCFC input **106**). In some embodiments, it is

assumed that a lower-voltage DC voltage is around 400 volts and that a higher-voltage DC voltage is around 800 volts. However, other voltage values may be used depending on (among other things) the voltage of the power supply **102** in the vehicle **100** and the voltage provided by a charging station **130**.

(24) In this particular example, the switches **114** can be used to selectively form and break an electrical connection between the AC input **104** and the AC/DC PFC converter **110**, and the switches **116** can be used to selectively form and break an electrical connection between the AC/DC PFC converter **110** and the DC/DC converter **112**. Also, the switches **118** can be used to selectively form and break an electrical connection between the DC/DC converter **112** and the power supply **102**, and the switches **120** can be used to selectively form and break an electrical connection between the DCFC input **106** and the power supply **102**. In addition, the switches **122** can be used to selectively form and break an electrical connection between the DCFC input **106** and the AC/DC PFC converter **110**, and the switches **124** can be used to selectively form and break an electrical connection between the AC/DC PFC converter **110** and the power supply **102**. Each of the switches **114-124** represents any suitable structure configured to selectively form and break an electrical connection, such as a MOSFET transistor, other transistor, contactor, circuit breaker, etc.

(25) In this example, the switches **114-124** can be controlled using at least one controller **126**. For example, the controller **126** may use inputs from one or more sensors to detect if and when the vehicle **100** is connected to an AC power source, a higher-voltage DC power source, or a lower-voltage DC power source. The controller **126** may use these inputs in order to determine which switches **114-124** should be opened and which switches **114-124** should be closed in order to support proper recharging of the power supply **102** based on the detected power source. The controller **126** includes any suitable structure configured to control the operation of electrical switches (and optionally other components) of the vehicle **100**. As particular examples, the controller **126** may represent at least one microprocessor, microcontroller, digital signal processor (DSP), application specific integrated circuit (ASIC), field programmable gate array (FPGA), or discrete circuitry.

(26) FIGS. **2** through **4** illustrate example configurations of the vehicle **100** during charging of the vehicle's power supply **102** using different sources of electrical power according to this disclosure. As shown in FIG. **2**, it is assumed here that the vehicle **100** is coupled to an AC power source **128** and is therefore receiving an AC electrical signal via the AC input **104**. In this configuration, the power supply **102** is charged using the incoming AC electrical signal, so the switches **114**, **116**, **118** are closed and the switches **120-124** are opened. This configuration allows the AC electrical signal to pass from the AC input **104** into the on-board charger **108**, where the AC/DC PFC converter **110** and the DC/DC converter **112** collectively operate to convert the AC electrical signal into a higher-voltage DC signal suitable for charging the power supply **102**.

(27) As shown in FIG. **3**, it is assumed here that the vehicle **100** is coupled to a higher-voltage DC power source (such as a higher-voltage charging station **130**) and is therefore receiving a higher-voltage DC electrical signal via the DCFC input **106**. In this configuration, the power supply **102** is charged using the incoming higher-voltage DC electrical signal, so the switches **120** are closed and the switches **114-118**, **122-124** are opened. This configuration allows the higher-voltage DC electrical signal to pass from the DCFC input **106** to the power supply **102** for charging of the power supply **102**.

(28) As shown in FIG. **4**, it is assumed here that the vehicle **100** is coupled to a lower-voltage DC power source (such as a lower-voltage charging station **130**) and is therefore receiving a lower-voltage DC electrical signal via the DCFC input **106**. In this configuration, the power supply **102** is charged using the incoming lower-voltage DC electrical signal, so the switches **114-120** are opened and the switches **122-124** are closed. This configuration allows the lower-voltage DC electrical signal to pass from the DCFC input **106** into the AC/DC PFC converter **110**, which is functioning in this configuration as a DC/DC boost circuit and which converts the lower-voltage DC electrical

signal into a higher-voltage DC electrical signal for charging the power supply **102**. Effectively, this configuration activates the front-end of the on-board charger **108** (the PFC converter **110**) when the vehicle **100** is coupled to a lower-voltage charging station **130** in order to boost the incoming DC voltage to a higher DC voltage that is suitable for charging the power supply **102**. The back-end of the on-board charger **108** (the DC/DC converter **112**) that provides isolation and voltage regulation may remain disconnected from the electrical pathway of the DC voltage when the vehicle is coupled to a lower-voltage charging station **130**, and the lower-voltage charging station **130** can have this isolation embedded internal to its architecture (so the isolation is not required from the vehicle side).

(29) As can be seen in FIGS. **2** through **4**, the power supply **102** in the vehicle **100** can be charged regardless of whether the vehicle **100** is connected to an AC power source, a higher-voltage DC power source, or a lower-voltage DC power source. Moreover, when the vehicle **100** is connected to a lower-voltage DC power source, the power supply **102** in the vehicle **100** can be charged by passing the incoming DC electrical signal through the PFC converter **110** of the on-board charger **108** (which is normally used only during AC charging) in order to increase the voltage level of the DC signal to a suitably-high voltage level. This allows for charging of the power supply **102** using both lower-voltage and higher-voltage charging stations **130** without requiring the use of a separate boost converter and without requiring the use of a motor winding of an electric motor of the vehicle **100**.

(30) Although FIG. **1** illustrates one example of a vehicle **100** supporting charging of the vehicle's higher-voltage power supply **102** using a lower-voltage charging station **130** and FIGS. **2** through **4** illustrate examples of configurations of the vehicle **100** during charging of the vehicle's power supply **102** using different sources of electrical power, various changes may be made to FIGS. **1** through **4**. For example, various components shown in FIGS. **1** through **4** may be combined, further subdivided, replicated, omitted, or rearranged and additional components may be added according to particular needs. Also, components in electrical circuits can often be positioned and electrically connected in various ways, so the specific electrical pathways and the associated switches **114-124** shown in FIGS. **1** through **4** are for illustration and explanation only. Other embodiments of the vehicle **100** may include fewer or more electrical pathways and fewer or more associated switches in order to support reconfiguration of the vehicle **100** based on the specific source of electrical energy being used to charge the power supply **102**. As a particular example of this, while pairs of electrical switches are shown here, certain components of the vehicle **100** may be coupled to a common ground connection. As a result, single electrical switches (rather than pairs of electrical switches) may be used to form and break electrical connections to or from those components since those components may remain permanently coupled to the common ground connection (meaning electrical switches may not be needed to selectively couple those components to ground).

(31) FIG. **5** illustrates a specific example implementation of a vehicle **100** supporting charging of the vehicle's higher-voltage power supply **102** using a lower-voltage charging station **130** according to this disclosure. Note that one or more components of the vehicle **100** shown in FIGS. **1** through **4** are omitted from FIG. **5** for clarity. As shown in FIG. **5**, the vehicle **100** includes the power supply **102**, such as a battery like an 800-volt battery or battery module. Voltage rails **502** and **504** are used to electrically couple other components of the vehicle **100** to the power supply **102**. The voltage rails **502** and **504** are often referred to as voltage bus bars.

(32) In this example, the other components of the vehicle **100** coupled to the voltage rails **502** and **504** include at least one inverter **506** coupled to at least one front motor (FM) **508** of the vehicle **100** and/or at least one inverter **510** coupled to at least one rear motor (RM) **512** of the vehicle **100**. Each motor **508**, **512** represents an electric motor configured to impart rotational motion to one or more wheels of the vehicle **100**. Unlike conventional internal combustion engine vehicles, electric vehicles can be designed with different numbers of electric motors at different locations of the electric vehicles. For instance, an electric vehicle may include a single electric motor driving its

front wheels, multiple electric motors separately driving its front wheels, a single electric motor driving its rear wheels, multiple electric motors separately driving its rear wheels, or any suitable combination thereof. In FIG. 5, it is assumed that both front and rear motors **508**, **512** and their associated inverters **506**, **510** are used, but this need not be the case. Each inverter **506**, **510** generally operates to convert a higher-voltage DC electrical signal from the power supply **102** into a suitable AC electrical signal for its associated motor **508**, **512**. In some cases, for instance, each inverter **506**, **510** may convert an 800-volt DC electrical signal into a three-phase AC electrical signal.

(33) The other components of the vehicle **100** coupled to the voltage rails **502** and **504** also include one or more components **514** that are designed to remain operational when the power supply **102** is being recharged (even if the vehicle **100** is coupled to a lower-voltage charging station **130**). For example, the one or more components **514** may include an HVAC compressor or other HVAC components and a positive temperature coefficient (PTC) heater or other heater. These components may be designed to operate using higher voltage levels and can remain coupled to the power supply **102** during recharging of the power supply **102**. Among other things, this can permit usage of the vehicle's air conditioning during summer and the vehicle's heater during winter, and the PTC heater can be used to heat the power supply **102** when the power supply **102** is charged during cold weather.

(34) The other components of the vehicle **100** coupled to the voltage rails **502** and **504** further include one or more components that may or may not remain operational when the power supply **102** is being recharged. Here, these components include at least one DC/DC converter **516**, and each DC/DC converter **516** may be coupled to at least one accessory load **518**. Each DC/DC converter **516** is configured to convert the DC voltage from the power supply **102** into a different DC voltage for the associated accessory load(s) **518**. As a particular example, each DC/DC converter **516** may be configured to convert an 800-volt DC electrical signal from the power supply **102** into a 12-volt DC electrical signal for the associated accessory load(s) **518**. The accessory loads **518** can represent various accessories of the vehicle **100**, such as a radio, entertainment system, navigation system, or other accessories.

(35) Although FIG. 5 illustrates one specific example of an implementation of a vehicle **100** supporting charging of the vehicle's higher-voltage power supply **102** using a lower-voltage charging station **130**, various changes may be made to FIG. 5. For example, various components shown in FIG. 5 may be combined, further subdivided, replicated, omitted, or rearranged and additional components may be added according to particular needs. As a particular example, the specific number of motors, inverters, converters, accessory loads, and other components of FIG. 5 can easily vary depending on the implementation.

(36) FIG. 6 illustrates an example control system **600** for controlling charging of a vehicle's power supply **102** according to this disclosure. For ease of explanation, the control system **600** shown in FIG. 6 is described as being used in the vehicle **100** of FIG. 1. As a particular example, the control system **600** may be used within or as part of the controller **126** in the vehicle **100**. However, the control system **600** may be used in any other suitable vehicle designed in accordance with the teachings of this disclosure.

(37) As shown in FIG. 6, the AC/DC PFC converter **110** in this example of the vehicle **100** is represented using a boost PFC converter circuit **602**, which includes input terminals **604** configured to receive an AC voltage ($V_{sub.AC}$) or a DC voltage ($V_{sub.DC}$) depending on how the power supply **102** is being recharged. The AC or DC voltage is provided to a rectifier **606**, such as four diodes connected in a square arrangement. The rectifier **606** outputs an input voltage ($V_{sub.IN}$), which represents the DC voltage $V_{sub.DC}$ or a rectified version of the AC voltage $V_{sub.AC}$. The input voltage $V_{sub.IN}$ is provided to an inductor **608**, which is coupled in series with a diode **610**. The diode **610** is coupled to an output capacitor **612**, which is coupled in parallel with a load **614** (such as the DC/DC converter **112** or the power supply **102**). A switch **616** (such as a MOSFET or

other transistor) is coupled to a node between the inductor **608** and the diode **610** and to ground. The switch **616** can be closed in order to charge the inductor **608** without having electrical current pass through the diode **610**, capacitor **612**, and load **614**. A pulse width modulation (PWM) or on-off controller **618** controls whether the switch **616** is conductive (closed) or non-conductive (opened) in order to control the operation of the boost PFC converter circuit **602**. A sense resistor **620** can be used to generate a sense voltage based on an electrical current flowing through the inductor **608** (I.sub.IN or I.sub.INDUCTOR), and the sense voltage is used as an input to the on-off controller **618**.

(38) The control system **600** shown in FIG. **6** can be used to control the boost PFC converter circuit **602** depending on whether the boost PFC converter circuit **602** is being used with an AC voltage V.sub.AC (during charging of the power supply **102** using an AC power source **128**) or a lower-voltage DC input voltage V.sub.DC (during charging of the power supply **102** using a lower-voltage charging station **130**). When used with an AC voltage V.sub.AC, the control system **600** can control the boost PEC converter circuit **602** to provide power factor correction. When used with a lower-voltage DC input voltage V.sub.DC, the control system **600** can control the boost PFC converter circuit **602** to provide voltage boosting of the lower-voltage DC input voltage V.sub.DC.

(39) In this example, the control system **600** includes two resistors **622** and **624** forming a first voltage divider, which is used to divide an output voltage (V.sub.OUT) being provided by the boost PFC converter circuit **602** to the load **614**. An operational amplifier **626**, a resistor **628**, and a capacitor **630** collectively form an error amplifier that outputs an error signal (V.sub.ERROR) based on a difference between the voltage from the first voltage divider and a reference voltage (V.sub.REF). The error signal V.sub.ERROR is used to control the boost PFC converter circuit **602** during power factor correction and during voltage boosting.

(40) To provide control for power factor correction, the control system **600** includes two resistors **632**, **634** forming a second voltage divider and two resistors **636**, **638** forming a third voltage divider. Each of the second and third voltage dividers is used to divide the input voltage V.sub.IN. The voltage generated by the second voltage divider may be defined as $K_{sub.1} \times V_{sub.IN}$, where $K_{sub.1}$ represents a value defined by the resistances of the resistors **632**, **634**. The voltage generated by the second voltage divider is provided to a square function **640**, which squares the voltage generated by the second voltage divider. The voltage generated by the square function **640** may be defined as $K_{sub.2} \times V_{sub.IN}^{sup.2}$, where $K_{sub.2}$ represents the value of $K_{sub.1}$ squared. The voltage generated by the square function **640** and the error signal V.sub.ERROR are provided to a division function **642**, which divides the error signal V.sub.ERROR by the voltage from the square function **640**. The voltage generated by the division function **642** may be defined as $K_{sub.3} \times V_{sub.ERROR} / V_{sub.IN}^{sup.2}$, where $K_{sub.3}$ represents the inverse of $K_{sub.2}$. The voltage generated by the third voltage divider may be defined as $K_{sub.4} \times V_{sub.IN}$, where $K_{sub.4}$ represents a value defined by the resistances of the resistors **636**, **638**. The voltage from the third voltage divider and the voltage from the division function **642** are provided to a multiplication function **644**, which multiplies the voltage from the third voltage divider and the voltage from the division function **642**. The voltage generated by the multiplication function **644** may be defined as $K_{sub.5} \times V_{sub.ERROR} / V_{sub.IN}$, where $K_{sub.5}$ represents the product of $K_{sub.3}$ and $K_{sub.4}$. The voltage generated by the multiplication function **644** is provided to the on-off controller **618** for use in controlling the operation of the switch **616**. For example, signals **646** represent how the on-off controller **618** can control the operation of the switch **616** so that the average current through the inductor **608** is substantially in phase with the voltage across the inductor **608** (indicating a good power factor). These operations allow the control system **600** to optimize the boost PFC converter circuit **602** for use in performing power factor correction. This represents one control loop that is used in the control system **600**.

(41) To provide control for voltage boosting, the control system **600** includes an adder **648** and a comparator **650**. The adder **648** combines a voltage measured at the sense resistor **620** with a

sawtooth voltage, meaning a voltage that linearly increases from a lowest level (such as 0 volts) to a maximum level before resetting back to the lowest level. The comparator **650** compares the output of the adder **648** with the error signal $V_{\text{sub.ERROR}}$ and generates a voltage that is provided to the on-off controller **618** for use in controlling the operation of the switch **616**. Here, the error signal $V_{\text{sub.ERROR}}$ is used as a threshold, and the comparator **650** operates to indicate when the output of the adder **648** exceeds the threshold. Signals **652** represent how the on-off controller **618** can control the operation of the switch **616** to provide voltage boosting when the vehicle **100** is being charged using a lower-voltage charging station **130**. Here, the $V_{\text{sub.ERROR}} 1$ and $V_{\text{sub.ERROR}} 2$ signals represent different values of the error signal $V_{\text{sub.ERROR}}$ (different error thresholds), where the output of the adder **648** is superimposed on these error thresholds. The PWM.sub.1 and PWM.sub.2 signals indicate how the output from the on-off controller **618** can vary, and it can be seen that the on-off controller **618** turns the switch **616** on for longer periods of time when the threshold (represented by $V_{\text{sub.ERROR}}$) is higher and for shorter periods of time when the threshold is lower. These operations allow the control system **600** to optimize the boost PFC converter circuit **602** for use in voltage boosting of a lower-voltage DC input signal. This represents another control loop that is used in the control system **600**.

(42) Note that many of the operations shown in FIG. **6** may be implemented using any suitable hardware or any suitable combination of hardware and software/firmware instructions. In some embodiments, for instance, some of the operations shown in FIG. **6** can be embodied as software or firmware instructions that are executed by one or more unitary or multi-core central processing units or other processing device(s). In other embodiments, these operations can be implemented using hardware components. In general, this disclosure is not limited to any particular physical implementation of the control system **600**.

(43) Although FIG. **6** illustrates one example of a control system **600** for controlling charging of a vehicle's power supply **102**, various changes may be made to FIG. **6**. For example, various components shown in FIG. **6** may be combined, further subdivided, replicated, omitted, or rearranged and additional components may be added, according to particular needs. Also, any other suitable control system may be used to control one or more operations of a vehicle **100**.

(44) FIG. **7** illustrates another specific example implementation of a vehicle **100** supporting charging of the vehicle's higher-voltage power supply **102** using a lower-voltage charging station **130** according to this disclosure. Note that certain components of the vehicle **100** shown in FIGS. **1** through **4** are omitted from FIG. **7** for clarity. As shown in FIG. **7**, the vehicle's power supply **102** can be charged using a DC power source **702**, which may represent a lower-voltage charging station **130**. The vehicle's power supply **102** may also be charged using a single-phase AC power source **704** or a three-phase AC power source **706**. A higher-voltage charging station **130** may also be used to charge the power supply **102**, but this usage is not represented in FIG. **7** since it does not involve the on-board charger **108**.

(45) In this example, a boost PFC converter circuit **708** may have a similar form as the boost PFC converter circuit **602**, but the boost PFC converter circuit **708** includes three paths. Each path includes an inductor **710** and a pair of switches **712a** (such as MOSFETs or other transistors). In each path, the switches **712a** can be used to selectively couple the inductor **710** of that path to one end of an output capacitor **714**. Additional switches **712b** can be used to selectively couple the single-phase AC power source **704** to one end of the output capacitor **714**.

(46) Various switches **S1-S11** are also shown in FIG. **7**. The switches **S1-S11** can be controlled (such as by the controller **126**) to reconfigure the boost PFC converter circuit **708** depending on the type of voltage source being used to charge the power supply **102**. Table **716** in FIG. **7** illustrates which of the switches **S1-S11** can be closed (check mark) and which of the switches **S1-S11** can be opened (X mark) depending on the type of voltage source being used to charge the power supply **102**. For example, switches **S1-S4**, **S8**, and **S10** can be closed and switches **S5-S7**, **S9**, and **S11** can be opened when the power supply **102** is being charged using the single-phase AC power source

704. This configuration of the switches **S1-S11** couples the single-phase AC power source **704** to all three paths of the boost PFC converter circuit **708** (via switches **S1-S3** and **S10**) and to a return electrical path (via switch **S4**). This configuration of the switches **S1-S11** also couples the output of the boost PFC converter circuit **708** to the DC/DC converter **112** (via switch **S8**).

(47) Switches **S5-58** can be closed and switches **S1-S4** and **S9-S11** can be opened when the power supply **102** is being charged using the three-phase AC power source **706**. This configuration of the switches **S1-S11** couples the three-phase AC power source **706** to all three paths of the boost PFC converter circuit **708** (via switches **S5-57**). This configuration of the switches **S1-S11** also couples the output of the boost PFC converter circuit **708** to the DC/DC converter **112** (via switch **S8**).

(48) Switches **S1-S3**, **S9**, and **S11** can be closed and switches **S4-S8** and **S10** can be opened when the power supply **102** is being charged using the DC power source **702** (such as a lower-voltage charging station **130**). This configuration of the switches **S1-S11** couples the DC power source **702** to all three paths of the boost PFC converter circuit **708** (via switches **S1-S3** and **S11**). This configuration of the switches **S1-S11** also couples the output of the boost PFC converter circuit **708** to the power supply **102** while bypassing the DC/DC converter **112** (via switch **S9**).

(49) Although FIG. 7 illustrates another specific example of an implementation of a vehicle **100** supporting charging of the vehicle's higher-voltage power supply **102** using a lower-voltage charging station **130**, various changes may be made to FIG. 7. For example, various components shown in FIG. 7 may be combined, further subdivided, replicated, omitted, or rearranged and additional components may be added according to particular needs. Also, a vehicle **100** need not support all of the various charging modes shown in FIG. 7, such as when the vehicle **100** supports one (but not both) of the AC charging techniques.

(50) FIG. 8 illustrates an example method **800** for charging a vehicle's power supply **102** according to this disclosure. For ease of explanation, the method **800** shown in FIG. 8 is described as being performed using the vehicle **100** of FIG. 1. However, the method **800** may be used by any other suitable vehicle designed in accordance with the teachings of this disclosure (including those shown in FIG. 5 or FIG. 7).

(51) As shown in FIG. 8, incoming electrical energy to recharge a vehicle's power supply is received at step **802**. This may include, for example, the vehicle **100** receiving an AC voltage via the AC input **104** or a higher-voltage DC or lower-voltage DC voltage via the DCFC input **106**. An operating mode to be used to charge the vehicle's power supply is selected at step **804**. This may include, for example, the controller **126** of the vehicle **100** determining whether the vehicle **100** is receiving the AC voltage, the higher-voltage DC voltage, or the lower-voltage DC voltage.

(52) If the operating mode to be used is determined to be lower-voltage DC (LV DC) mode at step **806**, the vehicle's power supply is charged using the lower-voltage DC voltage at step **808**. This may include, for example, the controller **126** controlling the switches **114-124** so that the AC input **104** is disconnected, the DCFC input **106** is electrically coupled to the PFC converter **110**, and the PFC converter **110** is electrically coupled to the power supply **102**. In this configuration, the lower-voltage DC voltage being received can be boosted using the PFC converter **110**, and the boosted voltage can be used to charge the power supply **102**.

(53) If the operating mode to be used is determined to be higher-voltage DC (HV DC) mode at step **810**, the vehicle's power supply is charged using the higher-voltage DC voltage at step **812**. This may include, for example, the controller **126** controlling the switches **114-124** so that the AC input **104** is disconnected and the DCFC input **106** is electrically coupled to the power supply **102**. In this configuration, the higher-voltage DC voltage being received can be used to charge the power supply **102**.

(54) Otherwise, the vehicle's power supply is charged using the AC voltage at step **814**. This may include, for example, the controller **126** controlling the switches **114-124** so that the AC input **104** is electrically coupled to the on-board charger **108** and the DCFC input **106** is disconnected. In this configuration, the AC voltage being received can be used to charge the power supply **102**. Note that

while not shown here, an additional determination could be made whether the AC voltage represents a single-phase or three-phase AC electrical signal, and switches (such as the switches **S1-S11**) can be controlled based on which type of AC electrical signal is being received.

(55) Charging of the vehicle's power supply is completed at step **816**. This may include, for example, the controller **126** determining whether the power supply **102** has been charged to its maximum capacity. If so, the controller **126** can disconnect the switches **114-124** so that additional electrical energy is not provided to the power supply **102**. Otherwise, the controller **126** may allow charging to continue until the power source is decoupled from the vehicle **100** or the power supply **102** is fully charged.

(56) Although FIG. **8** illustrates one example of a method **800** for charging a vehicle's power supply **102**, various changes may be made to FIG. **8**. For example, while shown as a series of steps, various steps in FIG. **8** may overlap, occur in parallel, occur in a different order, or occur any number of times.

(57) The above description has often described using the AC/DC PFC converter **110** in a vehicle **100** in order to allow for boosting of a lower-voltage DC input signal received from a lower-voltage charging station **130**. However, it should be noted here that the AC/DC PFC converter **110** in the vehicle **100** should be designed to handle the increased electrical power that is associated with coupling the vehicle **100** to a lower-voltage charging station **130**. For example, as noted above, the AC/DC PFC converter **110** is typically used to convert an AC electrical signal into a DC electrical signal. Some AC/DC PFC converters **110** may be designed only to handle a relatively low amount of electrical power, such as a power of about 22 kilowatts. Depending on the design of the lower-voltage charging station **130**, the lower-voltage charging station **130** may theoretically deliver a larger amount of power, such as between 150 kilowatts and 200 kilowatts or even more. As a result, the AC/DC PFC converter **110** in the vehicle **100** should be designed to handle a larger amount of electrical power than would ordinarily be obtained from an AC power source. Having said that, it is routine for a DC charging station to output less than the theoretical maximum amount of power during charging. For instance, some lower-voltage charging stations **130** may only output around 55 kilowatts of power, even though they may be designed to provide between 150 kilowatts and 200 kilowatts. Because of this, the AC/DC PFC converter **110** in the vehicle **100** can be designed to handle a larger amount of electrical power, but not necessarily the theoretical maximum amount of power that might be obtainable from a charging station.

(58) In some embodiments, various functions described in this patent document are implemented or supported using machine-readable instructions that are stored on a non-transitory machine-readable medium. The phrase "machine-readable instructions" includes any type of instructions, including source code, object code, and executable code. The phrase "non-transitory machine-readable medium" includes any type of medium capable of being accessed by one or more processing devices or other devices, such as a read only memory (ROM), a random access memory (RAM), a Flash memory, a hard disk drive (HDD), or any other type of memory. A "non-transitory" medium excludes wired, wireless, optical, or other communication links that transport transitory electrical or other signals. Non-transitory media include media where data can be permanently stored and media where data can be stored and later overwritten.

(59) It may be advantageous to set forth definitions of certain words and phrases used throughout this patent document. The terms "include" and "comprise," as well as derivatives thereof, mean inclusion without limitation. The term "or" is inclusive, meaning and/or. The phrase "associated with," as well as derivatives thereof, may mean to include, be included within, interconnect with, contain, be contained within, connect to or with, couple to or with, be communicable with, cooperate with, interleave, juxtapose, be proximate to, be bound to or with, have, have a property of, have a relationship to or with, or the like. The phrase "at least one of," when used with a list of items, means that different combinations of one or more of the listed items may be used, and only one item in the list may be needed. For example, "at least one of: A, B, and C" includes any of the

following combinations: A, B, C, A and B, A and C, B and C, and A and B and C.

(60) The description in the present application should not be read as implying that any particular element, step, or function is an essential or critical element that must be included in the claim scope. The scope of patented subject matter is defined only by the allowed claims. Moreover, none of the claims invokes 35 U.S.C. § 112(f) with respect, to any of the appended claims or claim elements unless the exact words “means for” or “step for” are explicitly used in the particular claim, followed by a participle phrase identifying a function. Use of terms such as (but not limited to) “mechanism,” “module,” “device,” “unit,” “component,” “element,” “member,” “apparatus,” “machine,” “system,” “processor,” or “controller” within a claim is understood and intended to refer to structures known to those skilled in the relevant art, as further modified or enhanced by the features of the claims themselves, and is not intended to invoke 35 U.S.C. § 112(f).

(61) While this disclosure has described certain embodiments and generally associated methods, alterations and permutations of these embodiments and methods will be apparent to those skilled in the art. Accordingly, the above description of example embodiments does not define or constrain this disclosure. Other changes, substitutions, and alterations are also possible without departing from the spirit and scope of this disclosure, as defined by the following claims.

Claims

1. An apparatus for charging a power supply, the apparatus comprising: an alternating current (AC) connector configured to receive an AC electrical signal from an AC power source; a direct current (DC) connector configured to receive a first DC electrical signal having a lower voltage from a lower-voltage DC power source and a second DC electrical signal having a higher voltage from a higher-voltage DC power source; an on-board charger configured to charge the power supply using the AC electrical signal, the on-board charger comprising an AC/DC power factor correction (PFC) converter and a DC/DC converter; switches configured to selectively form and break electrical connections between the AC connector, the DC connector, the AC/DC PFC converter, the DC/DC converter, and the power supply; a controller configured to adjust the electrical connections formed by the switches; and one or more sensors configured to detect whether the AC connector is connected to the AC power source, the DC connector is connected to the lower-voltage DC power source, or the DC connector is connected to the higher-voltage DC power source, wherein the controller is configured to, in response to a detection using the one or more sensors that the DC connector is connected to the lower-voltage DC power source, place the switches in a first configuration to couple the DC connector to the AC/DC PFC converter and couple the AC/DC PFC converter to the power supply, and wherein the controller is configured to, in response to a detection using the one or more sensors that the DC connector is connected to the higher-voltage DC power source, place the switches in a second configuration to couple the DC connector to the power supply.
2. The apparatus of claim 1, wherein the switches in the first configuration bypass the DC/DC converter so that electrical energy flows from the AC/DC PFC converter without passing through the DC/DC converter.
3. The apparatus of claim 1, wherein the controller is configured to, in response to a detection using the one or more sensors that the AC connector is connected to the AC power source, place the switches in a third configuration to couple the AC connector to the AC/DC PFC converter, couple the AC/DC PFC converter to the DC/DC converter, and couple the DC/DC converter to the power supply.
4. The apparatus of claim 1, wherein: the AC/DC PFC converter comprises a boost PFC converter circuit, the boost PFC converter circuit comprising an inductor, an additional switch configured to selectively couple the inductor to ground, and an on-off controller configured to control the additional switch; and the apparatus further comprises: a first control loop configured, when the AC

connector receives the AC electrical signal, to provide a first voltage to the on-off controller in order to control a power factor of the boost PFC converter circuit; and a second control loop configured, when the DC connector receives the first DC electrical signal having the lower voltage, to provide a second voltage to the on-off controller in order to control voltage boosting by the boost PFC converter circuit.

5. The apparatus of claim 1, wherein the controller is configured to control the switches based on whether the AC electrical signal comprises a single-phase AC electrical signal or a three-phase AC electrical signal.

6. The apparatus of claim 1, wherein: the power supply has a nominal voltage of approximately 800 volts; the lower-voltage DC power source is a 400-volt charging station; the higher-voltage DC power source is an 800-volt charging station; and the controller is configured to, using the one or more sensors, adjust the electrical connections formed by the switches such that (i) the DC connector is coupled to the power supply when the DC connector is coupled to the 800-volt charging station and (ii) the DC connector is coupled to the AC/DC PFC converter and the AC/DC PFC converter is coupled to the power supply when the DC connector is coupled to the 400-volt charging station.

7. An electric vehicle comprising: at least one electric motor; a power supply configured to provide electrical energy to the at least one electric motor; an alternating current (AC) connector configured to receive an AC electrical signal from an AC power source; a direct current (DC) connector configured to receive a first DC electrical signal having a lower voltage from a lower-voltage DC power source and a second DC electrical signal having a higher voltage from a higher-voltage DC power source; an on-board charger configured to charge the power supply using the AC electrical signal, the on-board charger comprising an AC/DC power factor correction (PFC) converter and a DC/DC converter; switches configured to selectively form and break electrical connections between the AC connector, the DC connector, the AC/DC PFC converter, the DC/DC converter, and the power supply; a controller configured to adjust the electrical connections formed by the switches; and one or more sensors configured to detect whether the AC connector is connected to the AC power source, the DC connector is connected to the lower-voltage DC power source, or the DC connector is connected to the higher-voltage DC power source, wherein the controller is configured to, in response to a detection using the one or more sensors that the DC connector is connected to the lower-voltage DC power source, place the switches in a first configuration to couple the DC connector to the AC/DC PFC converter and couple the AC/DC PFC converter to the power supply, and wherein the controller is configured to, in response to a detection using the one or more sensors that the DC connector is connected to the higher-voltage DC power source, place the switches in a second configuration to couple the DC connector to the power supply.

8. The electric vehicle of claim 7, wherein the switches in the first configuration bypass the DC/DC converter when the first DC electrical signal is used to charge the power supply.

9. The electric vehicle of claim 7, wherein the controller is configured to, in response to a detection using the one or more sensors that the AC connector is connected to the AC power source, place the switches in a third configuration to couple the AC connector to the AC/DC PFC converter, couple the AC/DC PFC converter to the DC/DC converter, and couple the DC/DC converter to the power supply.

10. The electric vehicle of claim 7, wherein: the AC/DC PFC converter comprises a boost PFC converter circuit, the boost PFC converter circuit comprising an inductor, an additional switch configured to selectively couple the inductor to ground, and an on-off controller configured to control the additional switch; and the electric vehicle further comprises: a first control loop configured, when the AC connector receives the AC electrical signal, to provide a first voltage to the on-off controller in order to control a power factor of the boost PFC converter circuit; and a second control loop configured, when the DC connector receives the first DC electrical signal having the lower voltage, to provide a second voltage to the on-off controller in order to control

voltage boosting by the boost PFC converter circuit.

11. The electric vehicle of claim 7, wherein the controller is configured to control the switches based on whether the AC electrical signal comprises a single-phase AC electrical signal or a three-phase AC electrical signal.

12. The electric vehicle of claim 7, wherein: the power supply has a nominal voltage of approximately 800 volts; the lower-voltage DC power source is a 400-volt charging station; the higher-voltage DC power source is an 800-volt charging station; and the controller is configured to, using the one or more sensors, adjust the electrical connections formed by the switches such that (i) the DC connector is coupled to the power supply when the DC connector is coupled to the 800-volt charging station and (ii) the DC connector is coupled to the AC/DC PFC converter and the AC/DC PFC converter is coupled to the power supply when the DC connector is coupled to the 400-volt charging station.

13. The electric vehicle of claim 7, further comprising: one or more first components configured to remain powered while the power supply is charged using any of the AC electrical signal, the first DC electrical signal having the lower voltage, and the second DC electrical signal having the higher voltage; and one or more second components configured to be selectively powered depending on whether the power supply is charged using the AC electrical signal, the first DC electrical signal having the lower voltage, or the second DC electrical signal having the higher voltage.

14. The electric vehicle of claim 13, wherein: the one or more first components comprise at least one heating, air conditioning, and ventilation (HVAC) compressor or heater; and the one or more second components comprise one or more vehicle accessories.

15. A method comprising: determining, using one or more sensors, whether an alternating current (AC) connector is connected to an AC power source to receive an AC electrical signal, a direct current (DC) connector is connected to a lower-voltage DC power source to receive a first DC electrical signal having a lower voltage, or the DC connector is connected to a higher-voltage DC power source to receive a second DC electrical signal having a higher voltage; and based on the determination using the one or more sensors, controlling switches to selectively form and break electrical connections between the AC connector, the DC connector, an AC/DC power factor correction (PFC) converter of an on-board charger, a DC/DC converter of the on-board charger, and a power supply, wherein controlling the switches comprises, in response to a determination using the one or more sensors that the DC connector is connected to the lower-voltage DC power source, placing the switches in a first configuration to couple the DC connector to the AC/DC PFC converter and couple the AC/DC PFC converter to the power supply, and wherein controlling the switches comprises, in response to a determination using the one or more sensors that the AC connector is connected to the AC power source, placing the switches in a third configuration to couple the AC connector to the AC/DC PFC converter, coupling the AC/DC PFC converter to the DC/DC converter, and coupling the DC/DC converter to the power supply.

16. The method of claim 15, wherein controlling the switches comprises, in response to a determination using the one or more sensors that the DC connector is connected to the higher-voltage DC power source, placing the switches in a second configuration to couple the DC connector to the power supply.

17. The method of claim 15, wherein: the power supply has a nominal voltage of approximately 800 volts; the lower-voltage DC power source is a 400-volt charging station; the higher-voltage DC power source is an 800-volt charging station; and the switches are controlled such that (i) the DC connector is coupled to the power supply when the DC connector is coupled to the 800-volt charging station and (ii) the DC connector is coupled to the AC/DC PFC converter and the AC/DC PFC converter is coupled to the power supply when the DC connector is coupled to the 400-volt charging station.
